Computer Science & DA

Warehousing

Lecture - 06

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Recap of Previous Lecture

Topic Data Smoothing

Topic Exponential Smoothing

Topic Data Normalization

Topics to be Covered

- **Topic** Data Normalization
- **Topic** Data Aggregation
- **Topic** Data Discretization

Topic : Exponential Smoothing

Week	Actual Sales	Smooth Value (Forecast) α=0.4
1	39	39
2	44	39
3	40	41
4	45	41
5	38	43
6	43	41
7	39	42
8		41

Topic : Exponential Smoothing

Week	Actual Sales	Smooth Value (Forecast) α=0.4
1	310	310
2	365	310
3	395	332
4	415	357
5	450	380
6	465	408
7	475	430
8		449

Topic : Data Normalization

Z-score Normalization

$$z_i = \frac{x_i - \mu}{\sigma}$$

$\mu = 17$

$\sigma = 3.6$

Values	$(x_i - \mu)$	Z	Mean of Z-score	SD of Z-score
12	-5	-1.39	0	1
14	-3	-0.83		
16	-1	-0.28		
18	1	0.28		
19	2	0.56		
23	6	1.67		

Mean of Z-score always zero

standard deviation of Z-score is always 1

Topic : Data Normalization

Z-score Normalization

- Z-score is measured in terms of standard deviations from the mean.
- If a Z-score is 0, it indicates that the data point's score is identical to the mean score.
- A Z-score of 1.0 would indicate a value that is one standard deviation from the mean.
- Z-scores may be positive or negative, with a positive value indicating the score is above the mean and a negative score indicating it is below the mean.

Topic : Data Normalization

Z-score Normalization with mean absolute deviation

$$z_i = \frac{x_i - \mu}{A}$$

$$A = \frac{\sum |x_i - \mu|}{N}$$

$$\sqrt{\frac{(x_i - \mu)^2}{N}}$$

Topic : Data Normalization

$\mu = 17$

Z-score Normalization with mean absolute deviation

$$z_i = \frac{x_i - \mu}{A}$$

$$A = \frac{\sum |x_i - \mu|}{N} = \frac{5+3+1+1+2+6}{6}$$

$$= 3$$

Values	$(x_i - \mu)$	Z
12	-5	-1.67
14	-3	-1
16	-1	-0.33
18	1	0.33
19	2	0.67
23	6	2

Topic : Data Normalization

Decimal Scaling Normalization

Find value of j such that:

$$max\left(\frac{x}{10^j}\right) \leq 1$$

56 → 0.056

138 → $\frac{138}{1000}$ = 0.138

24 → 0.024

Topic : Data Normalization

Decimal Scaling Normalization

Values	Normalized values
200	0.2
300	0.3
500	0.5
250	0.25
600	0.6
1000	1
700	0.7

$$max\left(\frac{x}{10^j}\right) \leq 1$$

for max value 1000

$$\frac{1000}{10^3} = 1$$

Topic : Data Normalization

Decimal Scaling Normalization

Values	
12	0.12
14	0.14
16	0.16
18	0.18
19	0.19
23	0.23
9	0.09

$$max\left(\frac{x}{10^j}\right) \leq 1$$

$j = 2$

Topic : Data Aggregation

The method of storing and presenting data in a summary format.

Year 2020

Year 2019

Year 2018

Quater	Cost
Q1	$224,000
Q2	$408,000
Q3	$350,000
Q4	$586,000

Year	Cost
2018	$1,568,000
2019	$2,356,000
2020	$3,594,000

Aggregated Data

Topic : Data Aggregation

Steps:

1. Data Grouping
2. Data Aggregation

Topic : Data Aggregation: Data Grouping

On Gender:

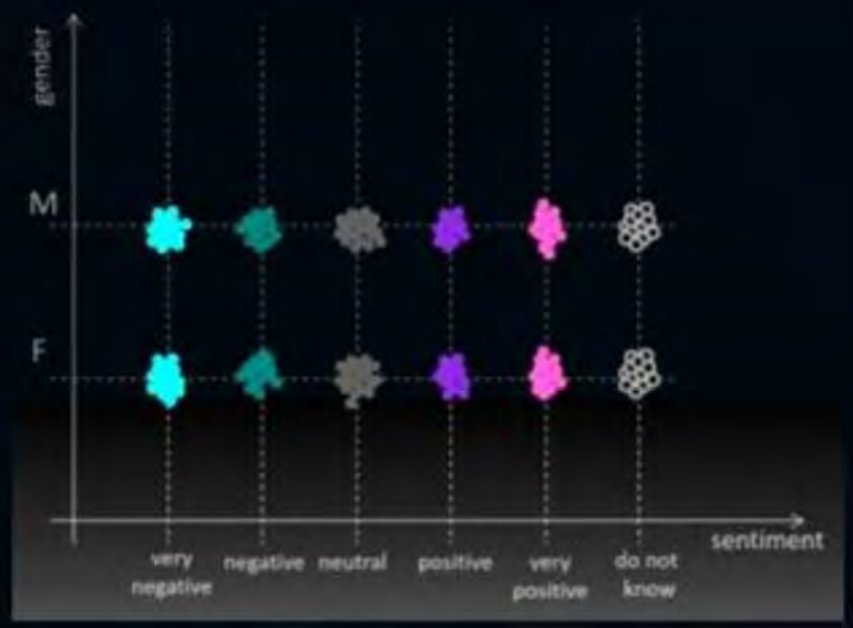
1. Male
2. Female

On Sentiments:

1. Very Negative
2. Negative
3. Average
4. Positive
5. Very Positive
6. Do not Know

On Gender & Sentiments:

1. Male, Very Negative
2. Male, Negative
3. Male, Average
4. Male, Positive
5. Male, Do not Know
6. Male, Very Positive
7. Female, Very Negative
8. Female, Negative
9. Female, Average
10. Female, Positive
11. Female, Very Positive
12. Female, Do not Know



Topic : Data Aggregation: Aggregation

Basic Measures:

- Count / Percent of rows
- Min/Max

Numerical:

- Sum
- Mean / Median
- Variance
- Kurtosis
- Skewedness

Date/Time:

- Number of days
- Min / Max

Topic : Data Discretization

A process of converting continuous data attribute values into a finite set of intervals with minimal loss of information.

Age	Age-groups
20	—
25	—
29	—
40	—
60	—
67	—
45	—
56	
⋮	

20-30	Young
31-40	Mature
41-50	Middle age
51-60	Senior
61-70	super —

Topic : Data Discretization

Why Data Discretization:

- Improves the quality of discovered knowledge
- Easy maintainability of the data
- Necessity to use discretized data by many DM algorithms

Data mining

- Reduces the running time of various data mining tasks such as association rule discovery, classification, and prediction
- Prepares data for further analysis, like classification
- Discretization is considered a data reduction mechanism because it diminishes data from a large domain of numeric values to a subset of categorical values

Topic : Data Discretization

Steps of Discretization

1. Sorting the continuous values of the feature to be discretized
2. Evaluating a cut point for splitting or adjacent intervals for merging
3. Splitting or merging intervals of continuous values according to some defined criterion
4. Stopping at some point

Topic : Data Discretization

Methods of Data Discretization

- Binning
- Histogram analysis
- Cluster analysis
- Decision tree analysis
- Correlation analysis

Topic : Data Discretization: Binning

- Top-down splitting technique
- Unsupervised discretization technique
- The main challenge is to choose the number of intervals or bins and how to decide on their width
- Binning methods smooth a sorted data value by consulting its “neighborhood”, that is the values around it. The sorted values are distributed into several “buckets” or bins. Because binning methods consult the neighborhood of values, they perform local smoothing.
- Attribute values can be discretized by applying equal-width or equal frequency binning, and then replacing each bin value by the bin mean or median.

Topic : Smoothing by Bin Means

8, 16, 9, 15, 21, 21, 24, 30, 26, 27, 30, 34

8, 9, 15, 16, 21, 21, 24, 26, 27, 30, 30, 34

Bins of 2 values
or
3 values
or
4 values

Bins on range => 8 to 15
16 to 23
24 to 31
32 to 39

Topic : Smoothing by Bin Boundary

8, 16, 9, 15, 21, 21, 24, 30, 26, 27, 30, 34

Topic : Data Discretization: Binning

Example:
Given a set of ages of people. Group them into 4 equal sizes of age ranges.
20, 24, 35, 23, 87, 56, 53, 43, 49, 90, 29, 35, 41, 92, 27, 31, 44

groups
1 – 25
26 – 50
51 – 75
76 – 100

Not a good method because we do not have values < 20 and > 92 ⟹ current range of values between 20 to 92

$$\frac{92 - 20}{4} = 18$$

⇒ groups

20 – 37
38 – 55
56 – 73
74 – 92

Topic : Data Discretization: Histogram Analysis

- Unsupervised discretization technique
- A histogram partitions the values of an attribute, into disjoint ranges, called buckets or bins

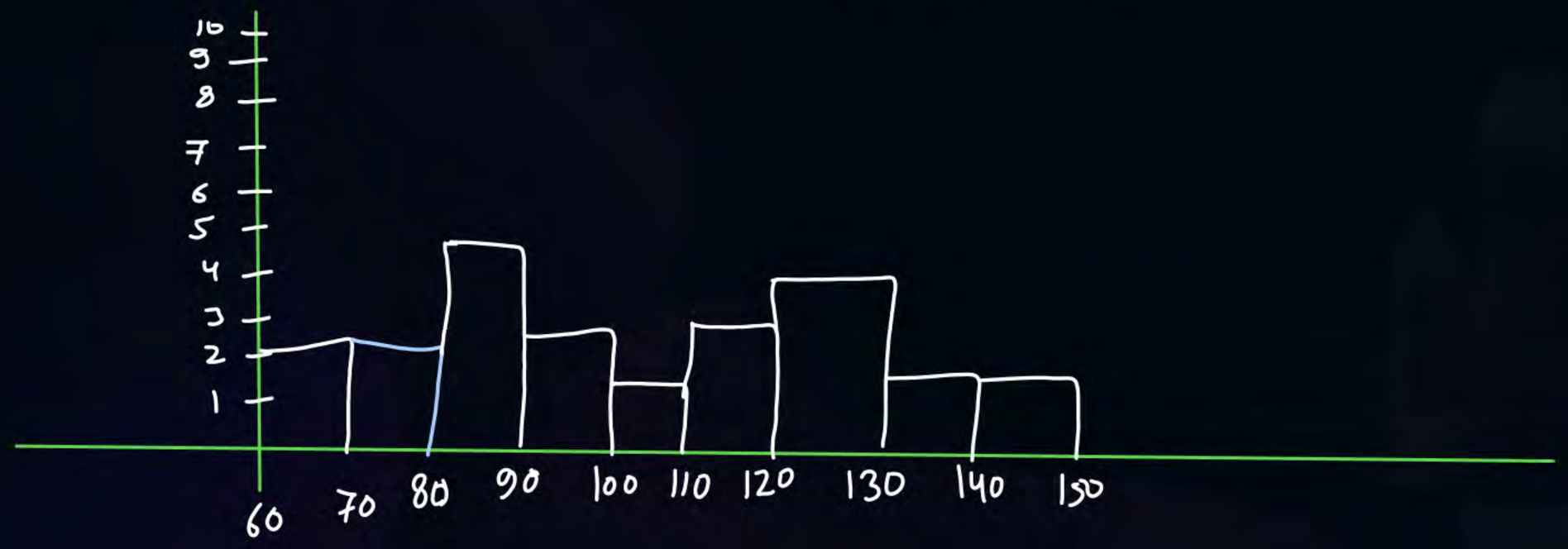
Topic : Data Discretization: Histogram Analysis

Topic : Data Discretization: Histogram Analysis

Consider the prices (in lacs) of properties sold in last one year in the Whitefield area of Bangalore:

85, 73, 81, 123, 112, 101, 126, 161, 65, 67, 89, 95, 77, 99, 111, 129, 142, 131, 84

Make a histogram of the prices in range of 10 Lacs ?



Topic : Data Discretization: Histogram Analysis

- Unsupervised discretization technique
- A histogram partitions the values of an attribute, into disjoint ranges, called buckets or bins
- If each bucket represents only a single attribute–value/frequency pair, the buckets are called singleton buckets. Singleton buckets are useful for storing high-frequency outliers

2 mins Summary

Topic Data Normalization

Topic Exponential Smoothing

Topic Data Discretization

Happy Learning

THANK - YOU